



SUREN NETWORK
PLAY WITH TECHNOLOGIES

DHCP CONFIGURATION

Step 1 : Assign ip address in all cisco routers

Step 2 : Add static route in R1 & R2

Step 3 : Enable dhcp server for HR department in R1

```
R1(config)# ip dhcp pool HR ( To create dhcp pool )
R1(config-dhcp)# network 50.0.0.0 255.0.0.0 ( To push ip & subnetmask )
R1(config-dhcp)# default-router 50.0.0.1 ( To push default gateway )
R1(config-dhcp)# dns-server 8.8.8.8 ( To push preferred dns )
R1(config-dhcp)# lease 1 2 3 ( To set validity )
R1(config-dhcp)#exit
R1(config)#
R1(config)# ip dhcp excluded-address 50.0.0.1 50.0.0.50 ( To set excluded )
R1(config)#exit
R1#
```

Step 4 : In vpc1 send a request to dhcp to get a ip
vpc-1> **ip dhcp** (To send a request to dhcp)

Step 5 : Check dhcp database

```
R1# show ip dhcp binding ( To check dhcp table )
```

Step 6 : Enable dhcp server for SALES department in same R1

```
R1(config)# ip dhcp pool SALES ( To create dhcp pool )
R1(config-dhcp)# network 192.168.2.0 255.0.0.0 ( To push ip & subnetmask )
R1(config-dhcp)# default-router 192.168.2.1 ( To push default gateway )
R1(config-dhcp)# dns-server 8.8.8.8 ( To push preferred dns )
R1(config-dhcp)# lease 1 2 3 ( To set validity )
R1(config-dhcp)#exit
R1(config)#
R1(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.50 ( To set excluded )
R1(config)#exit
R1#
```

Step 7 : Enable dhcp relay agent on router 2 client connected interface

```
R2(config)# interface ethernet 0/0  
R2(config)# ip helper-address 172.16.0.1      ( To enable relay agent )
```

Step 8 : In vpc 3 send a request to dhcp to get a ip
vpc-1> ip dhcp (To send a request to dhcp)

